Edbert Taidy

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LinkedIn | Portfolio | GitHub

CAREER SUMMARY

Graduate Software Engineer with hands-on experience designing, building, and deploying full-stack applications using **React, Vue.js, Angular, TypeScript, Java, Python, Spring Boot, NestJS, Flask, FastAPI, Django, PostgreSQL, and MongoDB**. Experienced in developing **RESTful APIs, AI-assisted classification pipelines, authentication and authorization mechanisms, database integrations, and CI/CD workflows** across independently developed projects. Strong foundation in **data structures and algorithms, object-oriented programming, secure software development, and web security**, supported by a Bachelor's degree in Computer Science (Digital Systems Security and Cyber Security), awarded with Distinction. Demonstrates the ability to take projects from concept to deployment, troubleshoot technical issues, and iteratively improve application functionality, reliability, and maintainability.

EDUCATION

Singapore Institute of Management-University of Wollongong Oct 2023 – Sept 2025
Bachelor of Computer Science Majoring in Digital Systems Security and Cyber Security (Double Major), Singapore

- Graduation Date: September 2025.
- Weighted Average Mark: 76.88 (Distinction).
- Achieved three high distinctions in **CSCI203 Algorithms and Data Structures, CSCI368 Network Security**, **CSIT328 Web Security**.

Singapore Institute of Management (SIM), Singapore Oct 2022 – Sept 2023
Diploma in Information Technology

PROJECTS

SignalBait — AI-Assisted Phishing & Scam Scanner Sep 2026 – Present
Vue.js, TypeScript, Tailwind CSS, Java, Spring Boot, Python, FastAPI, PostgreSQL, Hugging Face Transformers

- Developed a full-stack web application that analyzes suspicious messages using an AI-assisted classification pipeline, supporting both text and image inputs.
- Integrated the third-party **facebook/bart-large-mnli** model through **Hugging Face Transformers** for **zero-shot text classification.**
- Built **REST APIs with Spring Boot and FastAPI** to connect the frontend with backend and machine learning services.
- Implemented ticket reporting and tracking workflows, **including email notifications for ticket creation and status updates** using **Brevo.**
- Deployed the frontend and backend services using **Vercel and Render**.

SignalBait Admin Dashboard — Administrative Ticket Management System (Self-Directed Project) Sep 2026 – Present
Angular, TypeScript, Django, Python, PostgreSQL, ECharts, Tailwind CSS

- Developed an administrative dashboard to **manage user reports, update ticket statuses, and monitor ticket activity**.
- Implemented **session-based authentication** to restrict dashboard access to administrators.
- **Built analytics visualizations using ECharts** and an audit-log interface to review administrative actions.
- **Integrated Angular frontend components with Django API endpoints and a shared PostgreSQL database** used by the main application.
- Deployed the frontend and backend as separate services using **Vercel and Render**.

Urgent or Is It? – Cybersecurity Awareness Game (Self-Directed Project) Aug 2026 – Present
React, TypeScript, Next.js, NestJS, MongoDB, Tailwind CSS, Vercel

- **Developed and deployed** an independent full-stack cybersecurity awareness game using **React, TypeScript, NestJS, and MongoDB**, featuring interactive phishing-detection scenarios and persistent user progression.
- **Designed and implemented RESTful APIs** with NestJS and Mongoose for user registration, authentication, profiles, and **per-user game-state persistence**.
- **Implemented secure authentication and authorization** using **JWT, bcrypt password hashing, protected API endpoints, and DTO-based input validation** with NestJS guards and class-validator.
- **Built automated CI/CD workflows** with **GitHub Actions**, integrating production builds and deployment while configuring frontend-to-backend API communication across separate services.

Portfolio Website (Self-Directed Project) Jul 2026 – Present
React, TypeScript, Next.js, Vercel

- **Built and deployed** a production portfolio platform with **Next.js, React, and TypeScript**, applying reusable component architecture to create maintainable and scalable UI.
- **Implemented four-language localization** (English, Mandarin, Japanese, and Indonesian), enabling dynamic language switching across portfolio content.
- **Developed responsive and theme-aware UI**, incorporating **dark mode** and adaptive layouts for consistent cross-device usability.

TreeVisual – Binary Tree Learning Tool (Self-Directed Project) Feb 2026 – Present
Python, PySide6

- Designed and developed a desktop application to visualize **binary tree traversals** with step-by-step simulation.
- Implemented interactive controls to demonstrate **preorder, inorder, and postorder** traversal logic.
- Built save/load functionality for structured tree data using **JSON**.
- Focused on **clean UI design** to improve conceptual understanding of **data structures**.
- Packaged and distributed the application with downloadable builds for demonstration.

Solo Participant | Nanyang Technological University x Base 3 Hackathon Aug 2025 – Aug 2025
Phish or Miss — A Phishing Awareness Game (React, Vite, Flask, AWS EC2)

- Designed and built a **full-stack web application MVP within 5 days** in a competitive hackathon environment
- Deployed application on **AWS EC2 with Elastic IP**, enabling reliable public access for live demonstration during hackathon
- Implemented frontend-backend integration using REST APIs
- Debugged deployment and runtime issues under time pressure to ensure stable demo delivery

Final Year Project Team Leader | SIM-UOW Jan 2025 – May 2025
AI-nteractive: Face Animated Learning Management System
(Flask, MongoDB, SadTalker API)

- Led a team of 5 to integrate **AI-driven video generation** into a custom LMS
- Optimized processing pipeline, achieving **~90% reduction in generation time** using ffmpeg
- Designed backend workflows and API integration for automated content generation
- Applied **Agile/Scrum practices** to manage sprints and coordinate development

Software Development Methodologies Project Team Leader Oct 2024 – Nov 2024
Car Loaning Management System (HTML, PHP, SQL)

- Led a team of 7 to develop a **full-stack web system** with authentication and loan calculation features
- Implemented **CI/CD practices** during development lifecycle
- Delivered a functional MVP within 5 weeks following agile principles

EXPERIENCE

PPIS - Peduli Kartini Community Outreach Event May 2025
Logistics Assistant | Perhimpunan Pelajar Indonesia di Singapura (PPIS)

- Executed logistics support by distributing hampers and refreshments, and assisting participants in accessing legal and mental health services
- Collaborated with a team of 8 to coordinate setup and logistics for a 70+ participant community event

CERTIFICATIONS

- **JLPT (Japanese Language Proficiency Test) N3 –** ***Awarded 27 Aug 2026***
- **AWS Certified Cloud Practitioner (CLF-C02) –** ***Awarded 7 Aug 2025***
- **ISC2 Certified in Cybersecurity (CC) –** ***Awarded 29 Jul 2025***

SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, C++, PL/SQL
- **Frontend:** React, Next.js, Vite, HTML, CSS, Tailwind, Angular, Vue
- **Backend:** NestJS, Flask, PHP, REST APIs, Django, FastAPI, Spring Boot
- **Databases:** MongoDB, MySQL,PostGreSQL
- **Cloud & DevOps:** AWS (EC2, S3, CloudFront, Elastic IP), Vercel, Railway, Git, GitHub Actions, CI/CD, HTTPS
- **Practices:** Data Structures & Algorithms, OOP, Secure Coding, OWASP Top 10, Agile/Scrum, Debugging & Testing